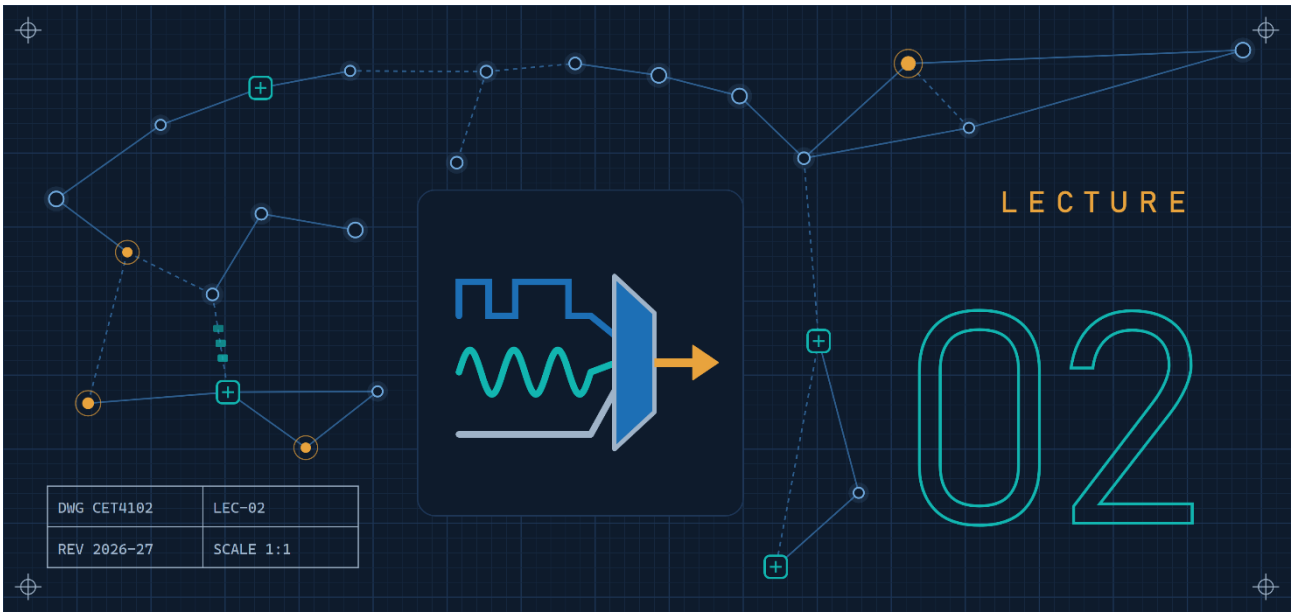
THE ISLAMIC UNIVERSITY
College of Technical Engineering
Department of Computer Techniques Engineering



LECTURE 02 · WEEK 2

Physical Layer Protocols, Transmission and Multiplexing

The functions that all protocols share, how the physical layer turns bits into signals, and how many users share one link.

Protocol functions	Physical layer and RS-232	Signals, data rate and media
Line coding and modulation	FDM, WDM and TDM	SONET/SDH and Bluetooth

Course	Computer Networks Protocols	**Code**	CET4102 · 6 ECTS
Stage	Fourth stage · Semester 7	**Session**	Theory lecture · Week 2
Instructor	Asst. Lect. Laith H. Alzubaidi	**Academic year**	2026 – 2027

Prepared for the fourth-stage students of the Department of Computer Techniques Engineering. Primary text: J. F. Kurose & K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Recommended: D. E. Comer, *Internetworking with TCP/IP*.

LEARNING OUTCOMES

By the end of this lecture you will be able to:

1. list the common protocol functions and give an example of each;
2. describe the physical layer and the four characteristics of an interface (RS-232);
3. relate bandwidth, baud rate and bit rate, and apply the Nyquist and Shannon formulas;
4. draw NRZ-L, NRZI and (differential) Manchester signals; tell ASK, FSK, PSK and QAM apart;
5. explain FDM, WDM and TDM, compute TDM rates, and describe SONET/SDH and Bluetooth.

Lecture 1 described the protocol stack from the top. This lecture starts at the bottom, with the **physical layer**: the only layer that touches the cable or the air. It turns the bits of a frame into a signal (a voltage, light or a radio wave) and back, without knowing what the bits mean. We first look at the jobs that all protocols share. Then we ask how bits become signals, what limits their speed, and how many users can share one link.

2.1 Protocol functions: a common framework

Lecture 1 split communication into layers, and each layer has its own protocol. It also listed design issues that every layer must solve, such as addressing and error control. Inside protocols at different layers, the same jobs appear again and again. These recurring jobs are called **protocol functions**. They give a framework for studying any protocol: ask which functions it performs, and how.

A small example shows several functions working together. A lower layer often limits the size of a PDU; for example, an Ethernet frame carries at most 1500 bytes of data. So the sender cuts a large message into numbered pieces: this is **segmentation**. It then adds a header and a trailer to each piece: this is **encapsulation**. The receiver checks each piece and puts the pieces back in order: this is **reassembly**.

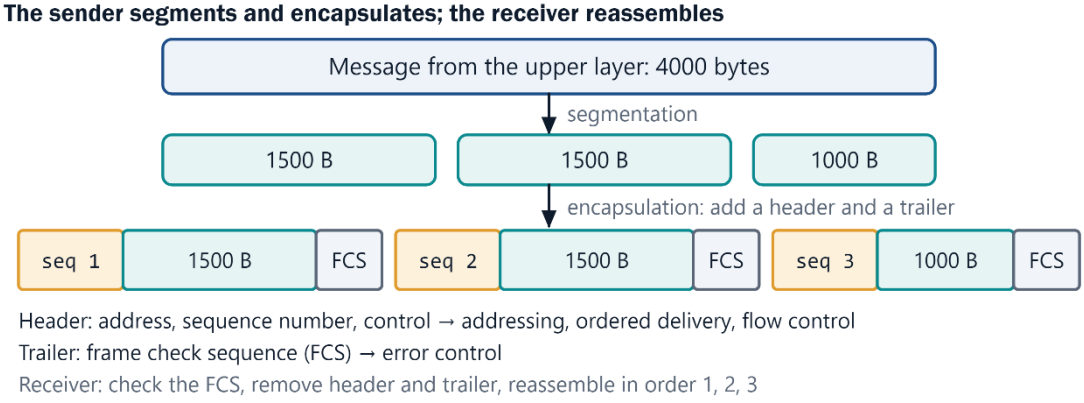


Figure 2.1 Segmentation, encapsulation and reassembly: the header fields and the trailer carry out several protocol functions.

In the figure, a 4000-byte message becomes pieces of 1500, 1500 and 1000 bytes with sequence numbers 1, 2 and 3. If piece 2 is damaged, only 1500 bytes are resent, not 4000. The cost is more header bytes to send and more PDUs to process.

The table lists all nine protocol functions, with an example of each.

Table 2.1 The protocol functions.

Function	What it does	Example
Encapsulation	Adds a header (and trailer) of control information	TCP, IP, Ethernet headers
Segmentation and reassembly	Splits data into small PDUs, then rebuilds it	TCP segments, IP fragments
Connection control	Sets up, uses and releases a connection	TCP handshake (Lecture 11)
Ordered delivery	Sequence numbers restore the right order	TCP, HDLC (Lecture 3)
Flow control	The receiver limits the sender's speed	Sliding window (Lecture 4)
Error control	Detects errors and has PDUs sent again	CRC (Lecture 5), ARQ (Lecture 4)
Addressing	Identifies the sender and the receiver	MAC, IP, port
Multiplexing	Lets several connections share one link	Ports; FDM, TDM
Transmission services	Priority, quality of service, security	IP priority, encryption

Multiplexing works in two directions. In **upward multiplexing**, several higher-level connections share one lower-level connection; for example, many TCP connections share one IP interface. In **downward multiplexing** (splitting), one connection is spread over several links for more capacity or reliability, as in link aggregation.

2.2 The physical layer: role, services and interface

The physical layer moves the individual bits of a frame from one node to the next over a transmission medium. It defines:

- **interface and medium:** the connector, the cable and the voltages (Section 2.2.1);
- **bit representation:** how 0 and 1 become signals (Section 2.4);
- **data rate:** bits per second, which fixes the duration of one bit;
- **bit synchronisation:** the receiver's clock must stay in step with the sender's;
- **line configuration and topology:** point-to-point or multipoint; star, bus, ring or mesh;
- **transmission mode:** *simplex* (one direction only, like a keyboard), *half duplex* (both directions, one at a time, like a walkie-talkie) or *full duplex* (both at once, like a phone call).

Physical-layer devices include the **repeater**, which regenerates a weak signal, the **hub** (a repeater with many ports) and the **modem** (modulator–demodulator).

2.2.1 The four characteristics of a physical interface

A physical-layer protocol (an *interface standard*) connects a **DTE** (data terminal equipment, such as a computer or router) to a **DCE** (data circuit-terminating equipment, such as a modem). It describes four characteristics, shown here for the classic serial interface **EIA-232** (RS-232).

Table 2.2 The four characteristics of a physical interface, with RS-232 values.

Characteristic	What it specifies	RS-232 example
Mechanical	Connector shape and number of pins	DB-25 or DE-9 connector

Characteristic	What it specifies	RS-232 example
Electrical	Voltage levels, timing, rate and distance	1 = −3 to −15 V, 0 = +3 to +15 V; up to 20 kbps over 15 m
Functional	The job of each pin	DE-9: pin 3 transmit, pin 2 receive, pin 5 ground
Procedural	The order of events used to send data	DTR/DSR (ready), RTS/CTS (permission to send), then data

RS-232 sends characters **asynchronously**. The idle line is at 1. Each character has a **start bit** (0), 8 data bits and a **stop bit** (1). This format is written **8N1** (8 data bits, no parity, 1 stop bit). So each character costs 10 bit times for 8 data bits: 80 % efficiency. A Cisco console port runs at 9600 bps, 8N1, so it sends 9600 ÷ 10 = 960 characters per second.

RS-232 suits only short, slow links. **RS-422** sends each signal on a pair of wires and reads their difference, so noise cancels. It reaches about 10 Mbps over 12 m, or 100 kbps over 1200 m.

2.3 Signals and transmission

2.3.1 Signals and bandwidth

Data must become a signal before it can travel. An **analog** signal can take any value within a range; a **digital** signal has only a few fixed levels. The simplest analog signal is the sine wave. It has a **peak amplitude** *A*, a **frequency** *f* in hertz (f = 1/T, where T is the period) and a **phase** (its position relative to time zero).

Real signals contain many frequencies. The bandwidth of a signal or channel is the width of its frequency range: $B = f_{high} - f_{low}$. A telephone voice channel passes about 300–3400 Hz, so its bandwidth is about 3.1 kHz.

2.3.2 Bit rate and baud rate

The **bit rate** *N* counts bits per second (bps). The **baud rate** *S* counts signal elements per second. If a signal has *L* levels, each element carries $\log_2 L$ bits:

$$N = S \times \log_2 L$$

For example, a 4-level signal at 1000 baud carries 1000 × 2 = 2000 bps. The figure below shows (a) the parts of a sine wave and (b) two digital signals with the same baud rate. Both send 4 elements in the same time, but the 4-level signal carries 2 bits per element, so its bit rate is twice as high.

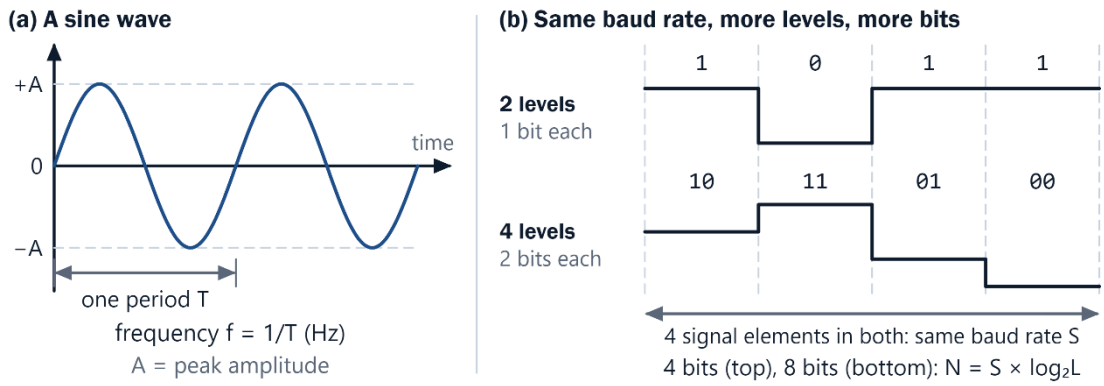


Figure 2.2 (a) A sine wave. (b) At the same baud rate, a 4-level signal carries twice as many bits.

COMMON MISTAKE

Bandwidth is measured in hertz, bit rate in bits per second and baud rate in signal elements per second. Bit rate equals baud rate only when L = 2.

2.3.3 Transmission impairments

The received signal is never exactly the sent signal. Three **impairments** change it:

- **Attenuation:** the signal loses energy with distance. Amplifiers or repeaters restore it.
- **Distortion:** the signal changes shape, because its frequency components travel at different speeds.
- **Noise:** unwanted energy is added: thermal noise, interference from motors, crosstalk from nearby wires and impulse spikes.

Power ratios are measured in **decibels**: $dB = 10 \log_{10}(P_2/P_1)$. A loss gives a negative value; −3 dB means half the power. Along a path, the dB values simply add. For example, a fibre with 0.2 dB/km loses 12 dB over 60 km, so 2 mW becomes $2 \times 10^{-1.2} \approx 0.13$ mW.

Noise is judged against the signal. The **signal-to-noise ratio** is $SNR = P_{signal} \div P_{noise}$, a plain number. In decibels, $SNR_{dB} = 10 \log_{10} SNR$, so 30 dB means SNR = 1000.

2.3.4 Maximum data rate of a channel

For a **noiseless** channel of bandwidth *B* with *L* levels, the **Nyquist** formula gives the maximum bit rate:

$$C = 2B \log_2 L$$

Real channels have noise, and levels that are close together are hard to tell apart. For a **noisy** channel, the **Shannon** formula gives the capacity, the highest rate that any coding can reach:

$$C = B \log_2(1 + SNR)$$

Here SNR is the plain ratio, not the dB value. In practice, Shannon gives the upper limit. We choose a rate below it, then use Nyquist to find the number of levels.

WORKED EXAMPLE 2.1: FROM SHANNON TO NYQUIST

A line has B = 4 kHz and SNR = 30 dB. (a) Find its capacity. (b) How many levels are needed for 32 kbps? (c) Could it carry 48 kbps?

1. $SNR = 10^{30/10} = 1000$.
2. Shannon: $C = 4000 \times \log_2(1001) = 4000 \times 9.97 \approx$ **39.9 kbps**.
3. 32 kbps is below the limit. Nyquist: $32\,000 = 2 \times 4000 \times \log_2 L$, so $\log_2 L = 4$ and **L = 16**.
4. The signal rate is 2B = 8000 baud, with 4 bits per element (16 signal levels).
5. 48 kbps is above 39.9 kbps, so **no** coding can carry it without errors.

2.3.5 Transmission media

Guided media keep the signal inside a cable. **Unguided media** send it through the air as electromagnetic waves. The table compares the main types.

Table 2.3 Guided and unguided transmission media.

Medium	Key idea	Typical use
Twisted pair	Two copper wires twisted together. Noise hits both wires equally, and the receiver reads their difference, so the noise cancels.	LANs up to 100 m (RJ-45, Cat 5e/6)
Coaxial cable	A centre conductor inside a metal shield	Cable TV, cable modems
Optical fibre	Light in a glass core; immune to electrical noise	Multimode: hundreds of metres; single-mode: tens of km
Radio (3 kHz–1 GHz)	Spreads in all directions; passes through walls	AM/FM radio, TV
Microwave (1–300 GHz)	Directional antennas focus it into line-of-sight beams	Relay towers, satellites; also mobile networks, Wi-Fi, Bluetooth
Infrared (300 GHz–400 THz)	Short range; blocked by walls	Remote controls

2.4 Encoding bits as signals

On a *baseband* link, such as a LAN cable, bits are sent directly as voltage levels: this is **line coding**. On a *band-pass* channel, such as a telephone line or radio band, bits must change a carrier wave: this is **modulation**.

2.4.1 Line coding

A good line code has enough **transitions** (level changes), because the receiver uses them to keep its clock in step. It should also have **no DC component** (no average voltage), because transformers in the path block DC. The figure below draws the four codes used in this course, and their rules follow it; some books use the opposite polarity.

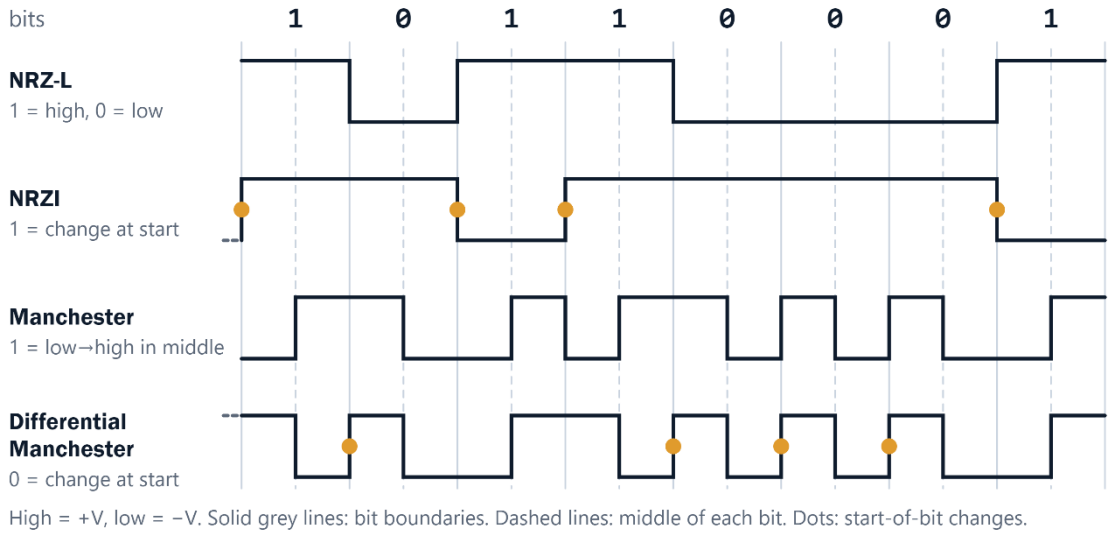


Figure 2.3 The bits 1 0 1 1 0 0 0 1 in four line codes (NRZI starts low; differential Manchester starts high).

- **NRZ-L**: the level shows the bit. 1 = high, 0 = low (RS-232 uses NRZ-L, but sends 1 as the negative level).
- **NRZI**: a 1 causes a transition at the start of the bit; a 0 causes none.
- **Manchester** (IEEE 802.3): a transition in the middle of every bit. 1 = low→high, 0 = high→low (10 Mbps Ethernet).
- **Differential Manchester**: a transition in the middle of every bit for the clock; a 0 also has a transition at the start of the bit (Token Ring).

In the figure, look at the run of three 0s. NRZ-L and NRZI stay flat there, so the receiver's clock may drift. Both Manchester codes still change in the middle of every bit, so they are **self-clocking** and have no DC component. The price is bandwidth: 10 Mbps Ethernet sends 20 million signal elements per second.

2.4.2 Digital modulation

Modulation changes one parameter of a **carrier**, a high-frequency sine wave. A **modem** does this at the sender and reverses it at the receiver, for example on a telephone line. Each signal element of a modulated carrier is called a **symbol**.

- **ASK** (amplitude shift keying): two amplitudes. On-off keying is used on optical fibre. ASK is sensitive to noise.
- **FSK** (frequency shift keying): two frequencies. It resists noise but needs more bandwidth.
- **PSK** (phase shift keying): different phases. BPSK uses 0° and 180°; QPSK uses four phases (2 bits per symbol).
- **QAM** (quadrature amplitude modulation): changes amplitude *and* phase. 16-QAM carries 4 bits per symbol, 64-QAM 6 bits.

The left side of the figure below sends the same bits with ASK, FSK and PSK.

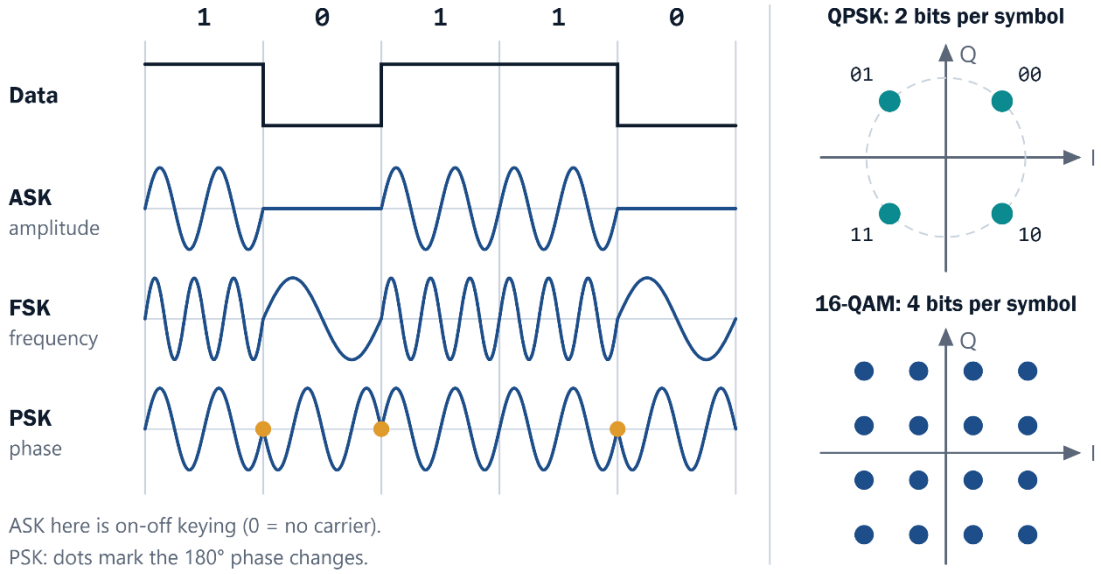


Figure 2.4 Left: the bits 1 0 1 1 0 sent with ASK, FSK and PSK. Right: constellation diagrams of QPSK and 16-QAM.

The right side shows **constellation diagrams**. Each dot is one symbol: its distance from the centre is the amplitude, and its angle is the phase. With M dots, bit rate = baud × $\log_2 M$. More dots give more bits, but they sit closer together and need a better SNR. For example, 2400 baud with 16-QAM gives 2400 × 4 = 9600 bps.

> **COMMON MISTAKE**
>
> AM, FM and PM are names for **analog** signals such as radio broadcasts. For **digital** data, use ASK, FSK, PSK and QAM.

2.5 Multiplexing

A fibre can carry far more than one user needs, so it pays to share it. **Multiplexing** lets n channels share one link. A **MUX** combines the inputs, and a **DEMUX** separates them at the other end. This is the physical-layer form of the multiplexing function of Section 2.1. (When independent stations share one broadcast channel without a central MUX, this is called multiple access: Lecture 3.)

2.5.1 FDM and WDM

In **FDM** (frequency-division multiplexing), each channel modulates its own carrier frequency, and the MUX adds the results into one signal. Unused **guard bands** keep neighbouring channels apart, and band-pass filters separate them at the receiver. Radio works this way: you choose a station by tuning to its frequency. For example, five 100 kHz channels with 10 kHz guard bands between them need at least 5 × 100 + 4 × 10 = 540 kHz.

WDM (wavelength-division multiplexing) is FDM for light: each channel uses its own wavelength (colour) in one fibre. Dense WDM carries dozens of wavelengths about 0.8 nm apart.

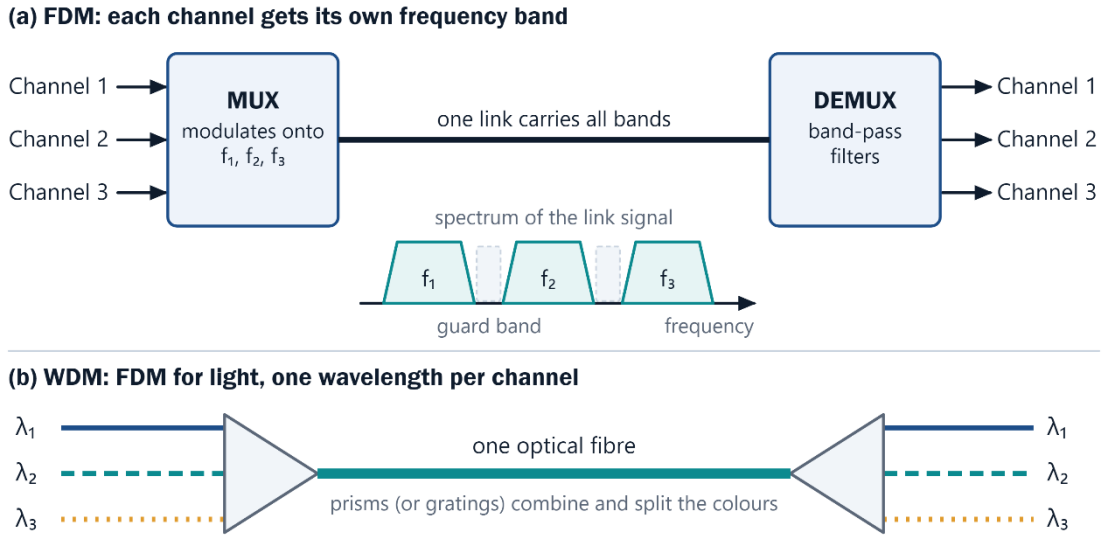


Figure 2.5 (a) FDM: one frequency band per channel, with guard bands between. (b) WDM: one wavelength per channel.

2.5.2 Synchronous TDM

In **TDM** (time-division multiplexing), the channels take turns using the whole link. In **synchronous TDM**, each **frame** has one fixed **slot** for every input. The slot position tells the DEMUX which input the data belongs to, and

a **framing bit** marks the start of each frame. The weakness is waste: an idle input's slot travels empty. For n inputs of rate R, u bits per slot and f framing bits per frame:

$$\text{frame size} = n \cdot u + f;\quad \text{frame rate} = R \div u;\quad \text{link rate} = \text{frame rate} \times \text{frame size}$$

WORKED EXAMPLE 2.2: SYNCHRONOUS TDM, T1 AND E1

Five terminals each send 1200 characters/s (8 bits each). Each frame takes one character per terminal and adds 1 framing bit. Find the frame size, frame rate, link rate and efficiency.

1. Frame size: 5 × 8 + 1 = **41 bits**.
2. Frame rate: **1200 frames/s** (one frame every 0.833 ms).
3. Link rate: 1200 × 41 = **49 200 bps** (48 000 bps of data + 1200 bps of framing).
4. Efficiency: 40 ÷ 41 ≈ **97.6 %**.

A voice channel (DS-0) is 8000 samples/s × 8 bits = 64 kbps. A **T1** frame carries 24 channels × 8 bits + 1 framing bit = 193 bits, sent 8000 times per second (every 125 µs): **1.544 Mbps**. An **E1** line (Europe and Iraq) has 32 slots × 8 bits × 8000 = **2.048 Mbps**: 30 voice slots, slot 0 for framing and slot 16 for signalling.

2.5.3 Statistical TDM

In **statistical TDM**, slots are given only to inputs that have data, so no slot is wasted. But the slot position no longer identifies the input, so each slot carries an **address** of $\lceil \log_2 N \rceil$ bits for N inputs. The link can be slower than the sum of the inputs, because they are rarely all busy; at peaks, data waits in a buffer.

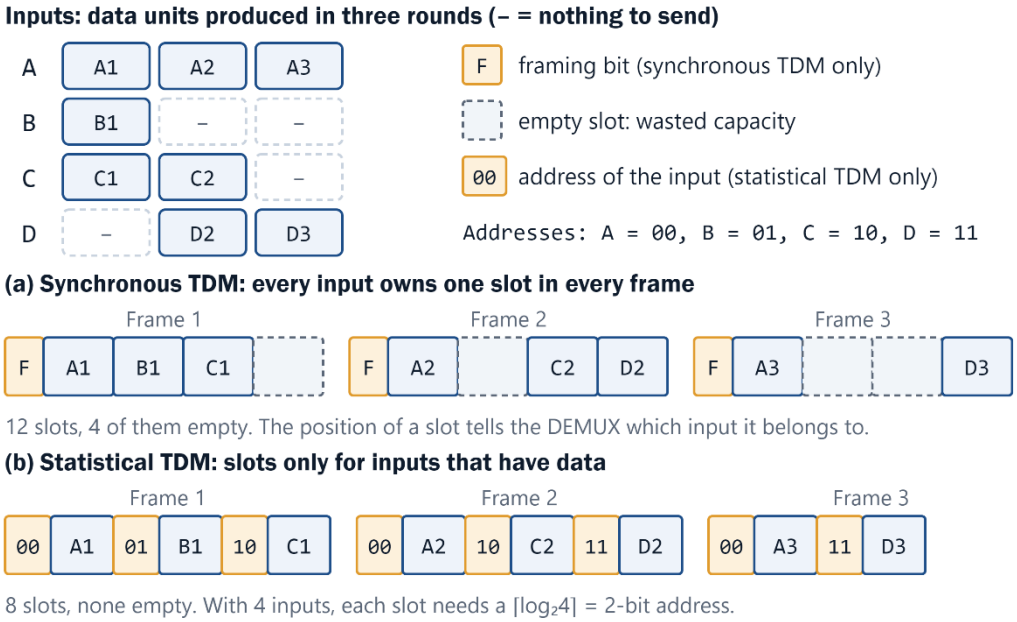


Figure 2.6 (a) Synchronous TDM: fixed slots and a framing bit. (b) Statistical TDM: addressed slots on demand.

For example, eight 100 kbps lines are each busy 25 % of the time. Synchronous TDM needs 800 kbps. Statistical TDM carries 200 kbps of data on average; with a 3-bit address per 8-bit slot it needs about 200 × 11/8 = 275 kbps, plus buffers.

2.6 Physical-layer protocols: SONET/SDH and Bluetooth

2.6.1 SONET/SDH

SONET (Synchronous Optical Network, North America) and **SDH** (Synchronous Digital Hierarchy, ITU-T, used in Europe and Iraq) are almost identical standards for synchronous TDM over fibre. Carriers use them as a **transport network**: a backbone that carries other traffic, such as E1 lines, IP and Ethernet. One master clock times the whole network, and a frame is sent every 125 µs (8000 frames/s).

A SONET network has three kinds of device: **STS multiplexers/demultiplexers**, where tributary streams enter and leave; **regenerators**, which restore weak signals; and **add/drop multiplexers** (ADM), which insert or remove some streams without unpacking the rest. A **section** joins two neighbouring devices, a **line** joins two multiplexers, and a **path** runs end to end. SONET has a layer for each of them (path, line and section), plus a **photonic** layer that sends the bits as light, like the OSI physical layer.

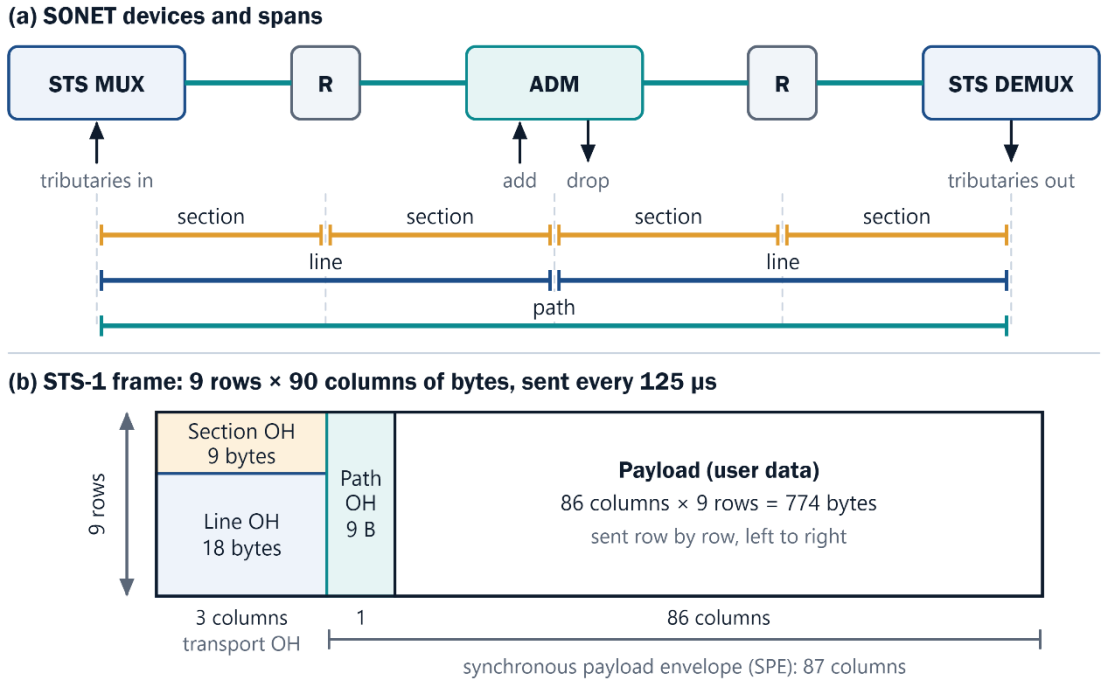


Figure 2.7 (a) SONET devices and the three kinds of span. (b) The STS-1 frame, 9 rows × 90 columns (not to scale).

The basic frame, **STS-1**, has 9 rows × 90 columns = 810 bytes and is sent row by row. The first 3 columns are transport overhead (section and line overhead). The other 87 columns form the payload envelope, and its first column is path overhead. The rate is 810 bytes × 8 bits × 8000 frames/s = **51.84 Mbps**. Each byte position repeats 8000 times per second, so it carries 64 kbps: exactly one voice channel. STS-*n* runs at *n* × 51.84 Mbps; OC-*n* is the same signal as light; SDH STM-*N* equals STS-3*N*.

Table 2.4 The main SONET and SDH rates.

SONET electrical	SONET optical	SDH	Rate (Mbps)
STS-1	OC-1	–	51.84
STS-3	OC-3	STM-1	155.52
STS-12	OC-12	STM-4	622.08

SONET electrical	SONET optical	SDH	Rate (Mbps)
STS-48	OC-48	STM-16	2488.32
STS-192	OC-192	STM-64	9953.28
STS-768	OC-768	STM-256	39 813.12

2.6.2 Bluetooth

Bluetooth (IEEE 802.15.1) is a **wireless personal area network**: it links devices a few metres apart, such as a phone and a headset. It uses 79 channels of 1 MHz in the 2.4 GHz band and hops to a new channel 1600 times per second, so each time slot lasts 625 µs. The basic rate is 1 Mbps.

Devices form a **piconet**: one **primary** and up to seven **active secondaries**, plus up to 255 **parked** secondaries (synchronised but inactive). A secondary sends only when the primary has addressed it. Piconets can be joined into a **scatternet** by a **bridge** device that takes turns between them. The bridge can be a secondary in both, or primary in one; it cannot be primary in two, because a piconet is defined by its primary.

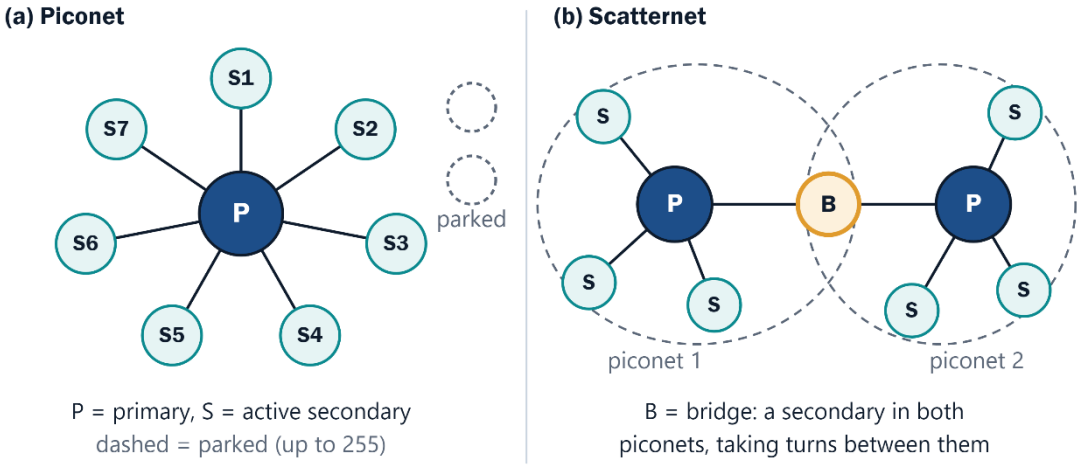


Figure 2.8 (a) Piconet: one primary, seven active and two parked secondaries. (b) Scatternet: two piconets, one bridge.

SUMMARY

- The same protocol functions appear at many layers: encapsulation, segmentation, connection control, ordered delivery, flow and error control, addressing, multiplexing and transmission services.
- A physical interface has mechanical, electrical, functional and procedural characteristics (RS-232 with 8N1: 10 bit times per character).
- $N = S \times \log_2 L$. Nyquist: $C = 2B \log_2 L$. Shannon: $C = B \log_2(1 + SNR)$. Signals suffer attenuation, distortion and noise.
- Manchester codes are self-clocking but need more bandwidth than NRZ. For ASK, FSK, PSK and QAM, bit rate = baud × $\log_2 M$.
- FDM and WDM share frequency. Synchronous TDM uses fixed slots (T1 = 1.544 Mbps, E1 = 2.048 Mbps); statistical TDM gives addressed slots on demand.
- SONET/SDH: 125 µs frames, STS-1 = 51.84 Mbps, STM-1 = 155.52 Mbps. Bluetooth: a piconet has 1 primary and up to 7 active secondaries; bridged piconets form a scatternet.

2.7 Review questions

1. List the nine protocol functions. Give two reasons for segmenting a large message, and one cost.
2. State the four characteristics of a physical interface, with the RS-232 value of each. How long does a 9600 bps, 8N1 console port take to send a 2400-character file?
3. (a) Explain the difference between bandwidth, baud rate and bit rate. (b) A signal with 8 levels is sent at 1000 baud; find its bit rate. (c) Name the three transmission impairments. Why does a twisted pair resist noise?
4. A channel has B = 20 kHz. (a) If it were noiseless, how many levels would carry 160 kbps, and at what baud rate? (b) Its SNR is 30 dB. Find its Shannon capacity. Is 160 kbps possible?
5. Draw NRZ-L, NRZI, Manchester and differential Manchester for 0 1 1 0 1 0 0 0. Before the first bit, the line is high for NRZI and low for differential Manchester. Which codes are self-clocking, and why?
6. What does each of ASK, FSK, PSK and QAM change? Find the bit rate of 64-QAM and of 16-QAM at 2400 baud.
7. Three 64 kbps voice channels are combined by synchronous TDM, one byte per channel per frame, plus 1 framing bit. Find the frame size, frame rate, frame duration, link rate and efficiency.
8. Give three differences between synchronous and statistical TDM. How many address bits are needed for 16 inputs?
9. Show that an STS-1 frame gives 51.84 Mbps. Find the STS-12 rate and its SDH name. Name the three SONET devices and the three kinds of span.
10. How many devices can be active in a Bluetooth piconet? What is a parked device? How is a scatternet formed?

2.8 References

- J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed., 2017: §1.2.2 (physical media), §1.3.2 (FDM and TDM).
- A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed., 2011: Chapter 2 (the physical layer), §4.6 (Bluetooth).
- B. A. Forouzan, *Data Communications and Networking*, 5th ed., 2013: Chapters 3–7, §14.3 (SONET), §15.3 (Bluetooth). Standards: EIA/TIA-232-F, ITU-T G.707, IEEE 802.15.1.
- Lecture material consulted: M. I. Shujaa (Middle Technical University); M. S. Mahmoud (Al-Mansour University College); L. A. Hassnawi (Al-Mustaqbal University); H. K. Ahmad (Al-Rafidain University).